

Ronin build manual.

Printing, build preparation and proposed assembly order / 04 October 2026

RONIN

Printed layers.
Borrowed silicon.
A sensible number of LEDs.

WORK IN PROGRESS

Build and remix this case freely, and make it your own. This project is changing rapidly: CAD, parts and build steps are still being revised. If you have questions, [reach out to Arthur](#). This draft is not a verified assembly guide.



Original SolidWorks-derived product render. Independent Art4Tech project; no official Voron affiliation. All manual artwork and prose are original.

Where each chapter starts.

Page numbers link to each chapter. Check what is confirmed and what still needs design or testing.

Read first / workflow	3-4
Current CAD locator views	5-6
BOM legend and all modeled parts	7-11
PCTG preparation, coupons and rear printing	12-15
Magnets, countersunk screws and inserts	16-17
LED locations and mounting concept	18-19
Cable-management planning and retention	20-21
LED harness reference and power adapters	22-23
Display: working behavior and roadmap	24
Open register / release checklist	25-28
Sources, credits and CAD provenance	29-30
Owner-proposed assembly order and hardware delta	31+

Check the parts before building.

Start with the current inventory, then resolve missing hardware and fit details.

BUILD + REMIX WELCOME

Build and remix this project freely. It is a rapidly evolving work in progress: CAD, the BOM and build steps may change. Check the current files and open issues as you go. If you have questions, [reach out to Arthur](#).

What is established

The export contains 83 active component instances from 34 unique CAD paths. Two Matchstick boards carry twenty RGB LEDs. Existing display firmware plays local animation/audio and receives serial resource values.

What remains open

Additional magnets, replacement countersunk screws, LED mounts, harnesses, USB cables, power-adaptor variants, cable retention and possible extra cooling. Final quantities and purchase part numbers are not yet complete.

- CAD quantity counts the instances in this export. Check the required quantity before buying; instance suffixes do not give the count.
- Read the open register before buying, printing a full set or wiring. A part number in the CAD filename may be a modeling reference.
- The owner-proposed order starts on page 31. Completed CAD is still needed for final exploded views and verified installation details. Diagrams are labeled as locators or concepts.

A magnet cannot proofread a BOM. Fortunately, we can.

Check one interface at a time.

Finish these design checks. The proposed physical assembly order begins on page 31.

01 / IDENTIFY

Check labels and purchased revisions against the quantities and supplier links.

02 / PRINT TRIAL

Dry PCTG and check the nozzle profile. Print clarity and fit coupons.

03 / FIT CAD

Resolve rear-cover split, magnets, inserts, LED mounts and optional fan.

04 / ROUTE

Model cable retention. Measure harness lengths with enough slack for service.

05 / VERIFY

Check connector mapping and power distribution against the exact boards.

06 / BUILD + TEST

Add final exploded pages, assemble, check covers/airflow, then repeat thermal tests.

The current assembly.

Locator view only - covers and internals retain the current CAD positions.



Printed shell

Untinted transparent PCTG is specified. Product renders use a frosted finish and 0.20 mm layer lines. Upright panels are preferred where practical.

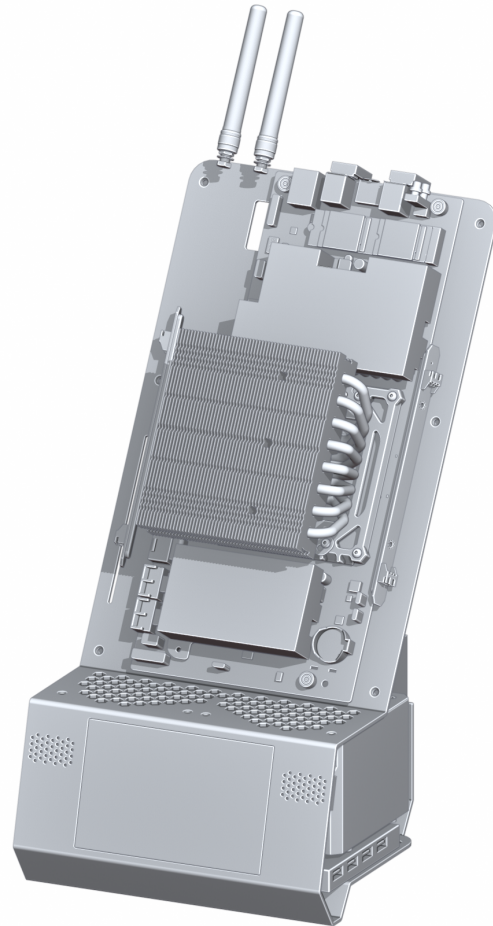
Still to design

LED positions are chosen. The mounts, extra magnet locations and cable-retention features still need CAD work.

Native CAD geometry. Diagram colors identify parts; they are not proposed filament colors.

Inside the covers.

Selected covers are hidden to show the current assembly. Components stay in their CAD positions.



Keep access clear

Leave access to the BC250 board, backbone, PSU, fans, storage and display. Allow room to unplug harnesses and remove and refit the magnetic covers.

Next CAD pass

Add the selected power boards, mounting hardware and cable routes before finalizing the assembly order.

The current export has four untessellated fan-face warnings. The figure record retains the source export hash. These warnings have not been repaired.

Count the models before ordering.

All 34 unique CAD paths are covered in the following four pages and editable CSV/JSON.

Field	Meaning
CAD qty	Active instances in this export: 83 in total. Additional or unmodeled items are listed separately.
Source PN	Part number in a CAD filename, where present. Confirm it before purchasing.
Purchase PN	Exact selected product/revision, supported by a label, order or supplier drawing. Still unconfirmed in this inventory.
Primary source	Official design or manufacturer reference. See editable BOM links and source appendix.
Open item	A choice to confirm before completing the BOM, buying hardware or writing the installation step.

Revision rule

When hardware or CAD changes, update the inventory, open register and affected figures. A similar model in the folder does not confirm the quantity or specification of a new part.

CAD inventory 1 / 4

CAD quantity is not the final required or purchased quantity. Source filenames are preserved in the editable inventory.

CAD source label	CAD qty	Class	Source PN	Release status	Sourcing
10x2 Magnet	6	bought	-	Exact purchased PN/revision open	Source TBD
16mm Power Button	1	bought	-	Exact purchased PN/revision open	Source TBD
180degree USB Adapter	2	bought	-	Exact purchased PN/revision open	Source TBD
6x3 Magnet	14	bought	-	Exact purchased PN/revision open	Source TBD
80mm Fan Grill	1	printed	-	Fabrication/specification open	Source TBD
90718A110 High-Strength Glass-Filled Nylon Plastic Washer	3	bought	90718A110	Exact purchased PN/revision open	Source ref(s)
92MM Arctic PST FAN	2	bought	-	Exact purchased PN/revision open	Source ref(s)
92MM Arctic PST Split Insert	2	printed	-	Fabrication/specification open	Source TBD
93657A009 Nylon Unthreaded Spacer	3	bought	93657A009	Exact purchased PN/revision open	Source ref(s)

Sourcing and replacements still need decisions. BC250_CAD_BOM_Rev_0p3.json retains CAD basenames, stable IDs, all instance names and source references.

CAD inventory 2 / 4

CAD quantity is not the final required or purchased quantity. Source filenames are preserved in the editable inventory.

CAD source label	CAD qty	Class	Source PN	Release status	Sourcing
93886A220 Reinforced Plastic Knurled-Head Thumb Nut	3	bought	93886A220	Exact purchased PN/revision open	Source ref(s)
94180A333 Brass Tapered Heat-Set Inserts for Plastic	8	bought	94180A333	Exact purchased PN/revision open	Source ref(s)
Apevia 500 SFX	1	bought	-	Exact purchased PN/revision open	Source TBD
Backbone	1	custom	-	Fabrication/specification open	Source TBD
bc250 after heatsink mod	1	custom	-	Fabrication/specification open	Source TBD
bc250 am4 adapter mount v1	1	custom	-	Fabrication/specification open	Source TBD
Dual 92MM FAN Shroud	1	printed	-	Fabrication/specification open	Source TBD
ESP32-P4-5 0-inch-20251230	1	bought	-	Exact purchased PN/revision open	Source ref(s)
Front Cover	1	printed	-	Fabrication/specification open	Source TBD

Sourcing and replacements still need decisions. BC250_CAD_BOM_Rev_0p3.json retains CAD basenames, stable IDs, all instance names and source references.

CAD inventory 3 / 4

CAD quantity is not the final required or purchased quantity. Source filenames are preserved in the editable inventory.

CAD source label	CAD qty	Class	Source PN	Release status	Sourcing
Left Apron	1	printed	-	Fabrication/specification open	Source TBD
M2 Splitter	1	bought	-	Exact purchased PN/revision open	Source TBD
M3x16 91294A134 Black-Oxide Alloy Steel Hex Drive Flat Head Screw	3	bought	91294A134	Exact purchased PN/revision open	Source ref(s)
M3x45 91420A138 Steel Phillips Flat Head Screws	8	bought	91420A138	Exact purchased PN/revision open	Source ref(s)
PanelMnt SMA Antenna Cable	2	bought	-	Exact purchased PN/revision open	Source TBD
rainbow-matchstick-diffuser	2	printed	-	Fabrication/specification open	Source TBD
Rainbow Matchstick	2	bought	-	Exact purchased PN/revision open	Source ref(s)
Rear Cover Whole	1	printed	-	Fabrication/specification open	Source TBD
Right Apron	1	printed	-	Fabrication/specification open	Source TBD

Sourcing and replacements still need decisions. BC250_CAD_BOM_Rev_0p3.json retains CAD basenames, stable IDs, all instance names and source references.

CAD inventory 4 / 4

CAD quantity is not the final required or purchased quantity. Source filenames are preserved in the editable inventory.

CAD source label	CAD qty	Class	Source PN	Release status	Sourcing
snowman mc75-8 heatsink v1	1	bought	-	Exact purchased PN/revision open	Source TBD
Speaker	2	bought	-	Exact purchased PN/revision open	Source TBD
SSD M2 2230	2	bought	-	Exact purchased PN/revision open	Source TBD
SSD M2 2280	1	bought	-	Exact purchased PN/revision open	Source TBD
Stubby SMA Wifi Antenna	2	bought	-	Exact purchased PN/revision open	Source TBD
Top Cover	1	printed	-	Fabrication/specification open	Source TBD
USB Hub	1	bought	-	Exact purchased PN/revision open	Source TBD

Sourcing and replacements still need decisions. BC250_CAD_BOM_Rev_0p3.json retains CAD basenames, stable IDs, all instance names and source references.

Twelve instances screened for fit.

Ten distinct part types were screened for printing. Confirm how each will be made before preparing a final slicer or G-code pack.

Instance	Intended-frame envelope / mm	Fit status
Rear_Cover_Whole-1	200.0 x 169.8 x 299.4	Rear: exceeds height
80mm_Fan_Grill-1	83.6 x 2.2 x 83.6	Geometric fit only
Dual_92MM_FAN_Shroud-1	194.2 x 100.4 x 70.0	Geometric fit only
92MM_Arctic_PST_Split_Insert-2	93.7 x 94.2 x 16.0	Geometric fit only
bc250_am4_adapter_mount_v1-1	107.0 x 67.0 x 96.3	Geometric fit only
92MM_Arctic_PST_Split_Insert-1	93.7 x 94.2 x 16.0	Geometric fit only
Front_Cover-1	200.0 x 166.4 x 193.4	Geometric fit only
Top_Cover-2	200.0 x 121.1 x 113.3	Geometric fit only
Left_Apron-1	100.0 x 194.0 x 106.4	Geometric fit only
Right_Apron-1	100.0 x 194.0 x 106.4	Geometric fit only
rainbow-matchstick-diffuser-2	4.7 x 80.2 x 131.0	Geometric fit only
rainbow-matchstick-diffuser-1	4.7 x 80.2 x 131.0	Geometric fit only

Dimensions come from unchanged mesh vertices. The top cover stands on its edge. The AM4 adapter is one CAD component with 19 bodies. Reconcile its physical parts and hardware before printing; review supports, brims and reliability in the slicer.

Trial clear PCTG before a panel.

Start with a small trial. These proposed settings have not been validated for a full panel.

First coupon trial	Starting point
Machine / material	CORE One+ INDX 8T / 3D-Fuel Natural/Clear Pro PCTG
Nozzle / layer / width	0.6 mm / 0.20 mm / 0.66 mm
Nozzle / textured PEI bed	260 C / 80 C
Outer / inner wall speed	15 / 20 mm/s; initial max volumetric speed 3 mm3/s
Cooling / flow	0% long-wall fan; test 0-10% locally. Calibrated 100% flow first.
Fill / seam	Avoid internal air gaps; aligned solid fill if present. Put seam on a less visible edge.

Prepare the material

Manufacturer drying guidance: 70 C for 6-8 h or 65 C for 8-12 h. Use the correct installed-nozzle INDX profile; the earlier active profile declared all tools 0.4 mm.

Check clarity and fit

Compare visibility and RGB diffusion. A readable sample that no longer fits its magnet pocket has not won the competition.

Measure haze. Measure fit.

The existing generic clarity coupons are 20 mm wide and 30 mm tall, in 1.5, 2.0 and 2.5 mm thicknesses.

- Choose the thickness nearest the panel wall. Print upright, then make a second sample rotated 90 degrees around Z.
- Compare 0.20 and 0.12 mm layers on the 0.6 nozzle. Test the 0.8 nozzle only if it still resolves the real features.
- Keep the best geometry settings. Compare 255 / 260 / 265 C, then calibrated 100 / 101 / 102% flow if dry samples show gaps. Change one variable at a time.
- Place the same text card 5 mm behind each coupon and compare under the same neutral light and LED color. Record thickness, haze, fit and settings.
- Print a small actual part before a panel. Check pockets, bridges and clips. The 0.4 nozzle remains useful for fine details.

Trial files

See the existing print_trials folder and PCTG_Print_Guide.html. The coupon geometries are generic test pieces. They do not replace case parts or validate a production print.

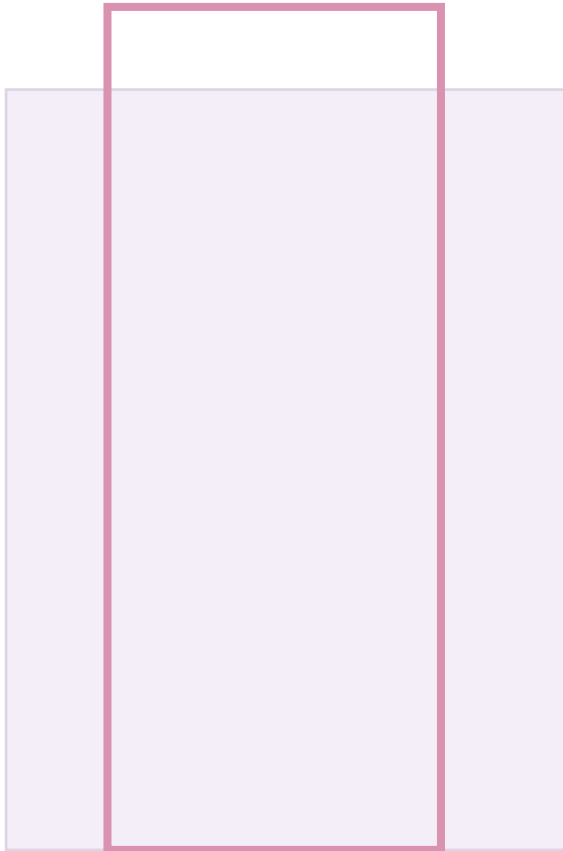
Keep the evidence

Photograph each labeled coupon and record its material/nozzle settings. Use a recipe for case parts only after actual print and fit checks.

The rear is taller than the printer.

A one-piece geometric fit exists with tilt, but the split decision remains open.

INDX height: 270 mm



Rear upright: 299.4 mm

Option A / one piece tilted

Verified unchanged-mesh envelope: 243.583 x 200.000 x 266.639 mm. Candidate transform: Rx(-19.2 degrees), then Rz(+90 degrees) in assembly axes. Import the supplied oriented trial as millimetres; do not copy those angles blindly into an arbitrary slicer.

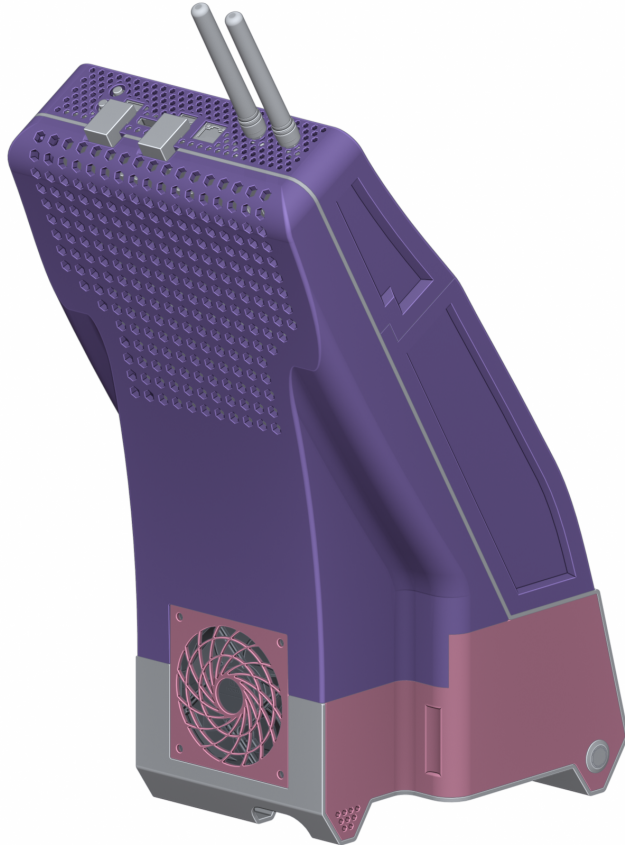
Option B / split for upright printing

A split can give more print margin and keep the layers upright, but adds CAD and assembly work. The split, seam placement and join hardware have not been designed.

The tilted object leaves about 2.21 / 2.50 mm per XY side and 3.36 mm top clearance. A 2 mm brim fits the object envelope geometrically; 3 mm does not. Supports and printer keep-outs remain unverified.

Check how the covers come off.

Use the current CAD count as a baseline. More magnets still need locations.



Current inventory

14 modeled 6 x 3 magnets; 6 modeled 10 x 2 magnets. Grade, coating, exact supplier PN, retention and final count remain open.

Before installing magnets

Pair and mark polarity on the actual build. Confirm pocket fit and retention with a small printed trial. Resolve the requested threaded magnets and their exact thread/body size.

Rear hexagonal vents are shown in their current geometry. Final cover-removal and magnet-location diagrams will use the revised CAD.

Choose the screw that fits the joint.

The current screws and inserts are modeling references. Replacements still need to be selected.

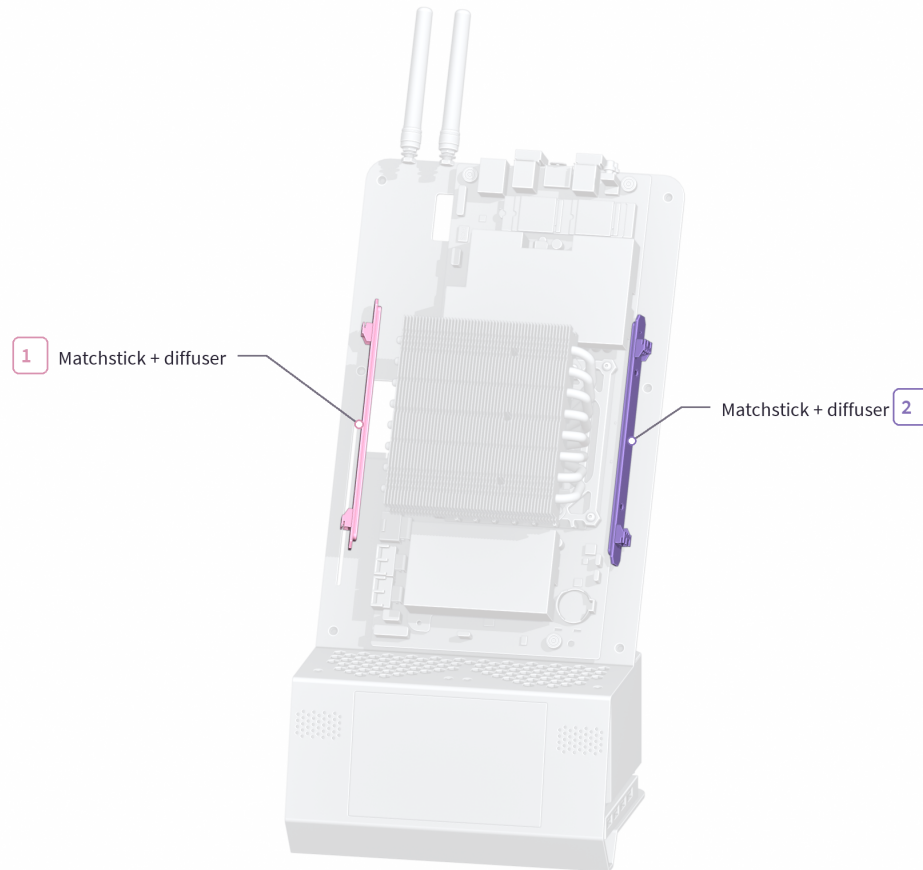
Modeled baseline	Decision needed
8 x M3x45; source label 91420A138	Select countersunk head standard, drive, head diameter/angle, length and engagement for each joint.
3 x M3x16; source label 91294A134	Confirm countersink geometry and replacement product, rather than copying the old filename.
8 x inserts; source label 94180A333	Confirm thread, OD, length, hole geometry and PCTG insertion procedure.
Washers / spacers / thumb nuts	Check the complete stack and mating hardware; CAD counts are retained in the BOM.

Check the joint before buying

Measure the countersink and assembled stack. Check full engagement without bottoming out. The head should sit flush without splitting the plastic and clear the electronics. Test the chosen insert in a coupon before setting installation temperatures and steps.

Twenty LEDs. Two chosen locations.

Positions come from the current assembly. Mounting remains a design task.



Technical ghost view — five outer covers hidden; neutral context faded.
Original positions shown. LED brackets and fasteners remain open.

Respect the optical path

The PCB and sheet-metal backbone occlude the emitters. Keep the useful glow around the board, through gaps and perforations, then through frosted PCTG.

No extra RGB fixtures

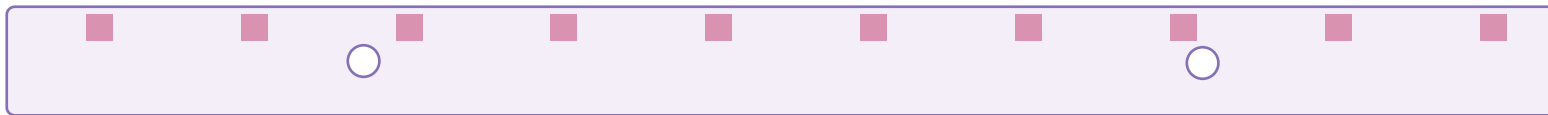
Only the two ten-LED Matchstick boards provide auxiliary RGB. The display backlight and fixed-white power-button indicator are separate.

Technical ghost view: five covers are hidden and nearby parts, including PCB/backbone, are faded to show occluded Matchsticks. CAD positions are unchanged. Numbers locate boards. The owner intends left first, right second; physical DIN/DOUT and connector orientation remain unverified. This view does not show physical transparency or measured light output.

Support the boards and leave access.

Original reference drawing and proposed bracket constraints. Check the actual purchased revision.

Reference outline: 158 x 11 mm



Two 3.2 mm plated mounting holes, tied to GND.
Reference center distance: approximately 85.00 mm.

Bracket concept to check

Try a removable printed bracket with insulated standoff clearance at the reference mounting holes. First confirm the actual hole locations, nearby geometry and appropriate M3 hardware.

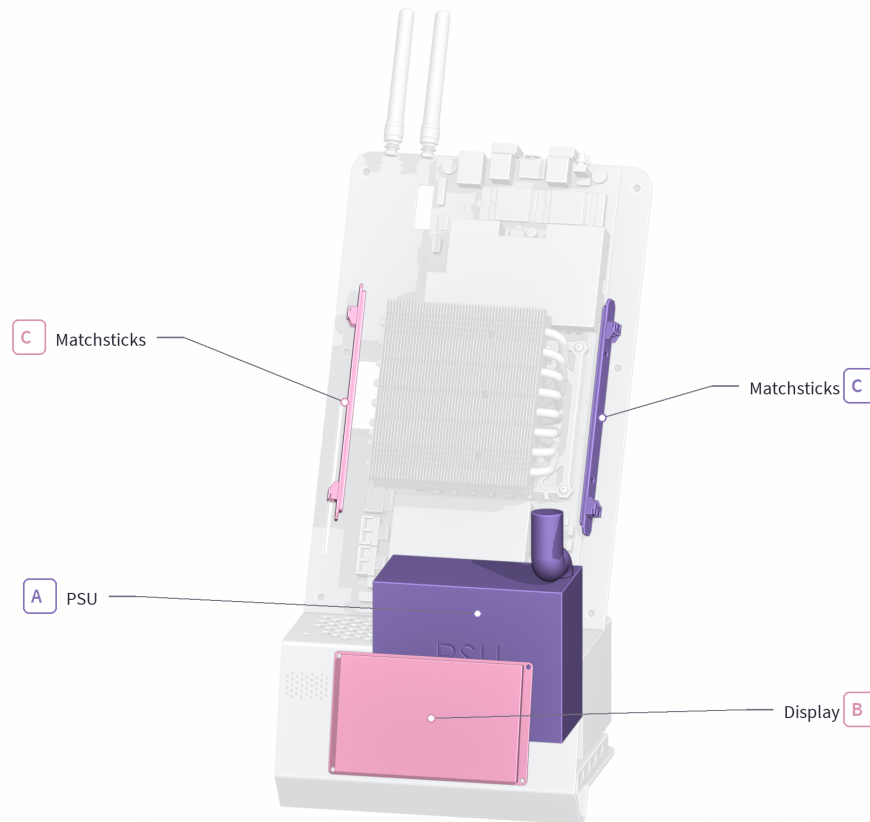
Alternative to compare

If screws are hard to reach, consider edge clips that leave the LEDs clear. Do not clamp LED faces or support boards by their cables. Keep connectors accessible and diffusers removable.

Board and LEDs are schematic. Native KiCad thickness is 1.6 mm; imported CAD PCB is 1.58 mm. No bracket has been modeled, collision-checked or physically validated.

Plan where spare cable will sit.

Three proposed routing regions. Anchor positions and hardware still need selection.



Technical ghost view — five outer covers hidden; neutral context faded.
Region labels only. Cable routes, tie points and LED mounting are unresolved.

A / PSU service loop

Bundle excess length near the PSU exit, clear of fans and vents. Leave it accessible for service.

B / display + USB + audio

Separate small cables from bulky power leads. Support connectors and leave the screen and covers removable.

C / LED perimeter harness

Plan the harness from the display to bar 1 to bar 2 along an accessible edge. Leave connector slack and keep wires off LED faces.

Technical ghost view: five covers are hidden and surrounding parts faded. A/B/C mark planning regions, not approved tie points or cable routes. Check the real harness, fan sweep and cover-removal path in revised CAD.

Service access beats a beautiful knot.

Measure the route before selecting mounts, cable ties, reusable straps and screws for the BOM.

Heavy PSU wiring

Compare tie saddles or reusable straps secured to checked backbone/bracket features. Support the bundle without using the cover as a structural anchor.

Small harnesses

Give USB, display and LED connectors separate strain relief. Match tie width and saddle shape to the bundle, and keep the bend radius of the selected cable.

- Plan access for unplugging and removing each cover before placing a tie saddle.
- Model the anchor body and fastener envelope. Check fan sweep, vent openings and nearby electronics.
- Mock up cable lengths with PSU excess and the bend direction of the right-angle connector. Leave enough slack for service, then secure the cables.
- Trial the fans, covers and service access before writing a numbered cable-routing page.

Do not assume a straight-through harness.

Official Matchstick design reference only. Verify the purchased boards and connector orientation before crimping.

Board interface	Reference pin assignment	Not yet established
J1 / input	1 DIN 2 +5V 3 GND	Exact purchased revision and physical pin-1 orientation
J2 / output	1 GND 2 +5V 3 DOUT	Correct housing, terminals and continuity mapping
J3 / alternate output	Single DOUT pad	Whether it is used in this build
Connector footprint	JST XH S3B-XH-A-1; nominal 2.50 mm	README says 2.54 mm; use actual connector drawing/revision

Chain intent

Display controller -> DIN of bar 1; DOUT of bar 1 -> DIN of bar 2.
Shared reference ground and the correct +5V LED supply are required. Reference J2/J1 pin order differs.

Firmware / power boundary

Display GPIO/header, logic-level compatibility, supply budget, protection, wire gauge, connector orientation and measured harness lengths remain open. This is not a connection-ready display pinout.

The adapter input is EPS, not PCIe.

Select the exact Mosfet.Part variants and add their mounting to CAD before writing final wiring pages.

EPS 8-pin to dual Micro-Fit

Official adapter takes a CPU EPS 8-pin input and distributes power to two Micro-Fit connections. The designer specifies use alongside the BC250 PCIe 8-pin input. Follow the exact product documentation; do not substitute a PCIe plug for EPS.

PS_ON adapter

Button, case-header, LITE and PSU-specific variants exist. Select the board for the actual PSU and case-button behavior. Model its mounting, terminals and harness before writing the installation step.

- Confirm actual PSU model, connector inventory and cable compatibility. Never infer a modular PSU cable pinout from connector appearance.
- Add both selected boards and any spacers, screws and cable restraints to the revised assembly and BOM.
- Document the main power-button indicator separately from the auxiliary RGB control path.

Exact product references are in the source appendix. No mains wiring or final power harness is designed in this edition.

What the display can do today.

The existing firmware works. The integrations below still need implementation.

Working today

Existing firmware plays local animated scenes with audio and receives CPU/memory/disk values over serial. An October 3 UART0 test showed live laptop values; LibreHardwareMonitor supplies Windows temperature. No new firmware was flashed for that test.

Next firmware work

Advanced twenty-pixel RGB scenes; resource-monitor options; context/page touch buttons that call host scripts; gamepad wake; and a voice-agent portal using the display mic/speakers.

The local companion prototype saves LED scenes, touch profiles and agent endpoint/model settings on the host. New device-control commands, script execution, voice integration and Windows/Linux installers still need implementation.

The promotional film animates genuine installed screen media with reconstructed source-defined GUI placeholders. Cinematic dim/sleep behavior is a render treatment, not a newly implemented firmware feature.

Before the next revision 1 / 3

Additional quantities and final part numbers remain open. Resolve these items in the next CAD pass.

Open item	Modeled baseline	Required confirmation
Additional 6 x 3 magnets	14	Additional quantity, grade, coating, polarity/orientation and retention fit.
Different M3 countersunk bolts	11	Length, head angle/diameter, drive, grade, quantity, thread engagement and countersink fit; existing model has 3 x16 and 8 x45.
Owner-requested 3 mm threaded magnets	0	Clarify M3 thread versus 3 mm body size, male/female thread, envelope, quantity, mounting and polarity.
Final heat-set inserts	8	Final PN, thread, OD/length, hole geometry, insertion method, quantity and pull-out test.
Display to LED bar 1 to LED bar 2 harness	0	Display output GPIO/header mapping, logic levels, +5V source/budget, common ground, LED revision pinout, connector housings/crimps, wire gauge, lengths, keyed orientation and test.
Mounts/fasteners/clips for two chosen LED positions	0	Support surfaces, hole alignment, insulated clearance, insert/bolt selection, connector access, diffuser retention and service removal; no mount validated.
USB extension cables	0	Connector genders, USB speed/power requirements, length, bend radius, panel retention and exact quantity.

Before the next revision 2 / 3

Additional quantities and final part numbers remain open. Resolve these items in the next CAD pass.

Open item	Modeled baseline	Required confirmation
90 degree USB-C to USB-A cable	0	Connector genders/orientation, data capability, cable length, bend clearance, strain relief and exact quantity; separate from two modeled 180-degree USB adapters.
Possible ARCTIC P12 Slim PWM PST	0	Selection/quantity, airflow direction, 120 x120 x15 mm clearance, hole pattern, fan power/header budget, wiring and mount validation.
Mosfet.Party EPS power distribution adapter	0	Confirm product/revision/quantity, PSU cable compatibility, board bounds, mounting, both BC250 power paths and harness retention.
Mosfet.Party PS_ON control adapter	0	Confirm ATX button/header versus LITE variant, board bounds/holes, button/LED connector pinout, PSU and BC250 sense wiring, quantity and mounting.
Excess PSU cable retention	0	Choose case anchor/tie-point locations, zip-tie/mount PN/quantity, routing, bend radius, insulation, fan clearance and service slack.
Small/USB/LED/audio cable retention	0	Choose separate anchor/tie-point locations, zip ties/mounts/retainers, lengths, strain relief, connector access and service slack.
Speaker wiring and final speaker parts	2	Exact speaker PN/impedance/rating, PH2.0 housing/pin orientation, harness length, mounting and clearance.

Before the next revision 3 / 3

Additional quantities and final part numbers remain open. Resolve these items in the next CAD pass.

Open item	Modeled baseline	Required confirmation
Wireless card and antenna identity	0	No explicitly named wireless card instance; confirm installed card PN, interface, adapter and antenna/bulkhead cable identities from owner evidence.
M.2 splitter identity	1	Manufacturer/PN, topology/interface compatibility, power, mounting and attached devices; generic CAD filename cannot establish exact purchased splitter.
Final SSD identities	3	PN/capacity/interface and role for 2 modeled 2230 and 1 modeled 2280 bodies.
USB hub identity	1	Exact PN, ports/speed, power source, cable and mounting.
PSU identity and cable set	1	Apevia 500 SFX filename only; confirm exact label PN/revision, connectors/pinouts and cable gauges.
Decompose the custom AM4 adapter subassembly	1	One imported custom CAD component contains 19 exported bodies. Identify fabricated parts and separately required hardware, checking against top-level washers, spacers and thumb nuts to avoid double-counting. Manufacturing/material and final physical quantities remain open; do not decompose bought board/PSU chips into separate purchases.
Other owner-mentioned missing components	None	Owner to identify remaining components; inventory is not a purchasing checklist.

Checks before final build instructions.

Confirm these items before treating the concepts as assembly instructions.

- Finalize hardware selection and source links. Compare required, purchased and modeled quantities. Resolve or remove placeholder parts.
- Update the SolidWorks assembly with extra magnets, threaded magnets, inserts, power boards, selected cable ends, LED mounts and cable retention.
- Check the rear printing strategy in the actual INDX slicer profile. Print and measure representative PCTG parts.
- Verify all harness pin mapping and chosen board revisions. Measure lengths with service slack and document strain relief.
- Generate original exploded views from that CAD revision. Write one mechanically verified operation per step, with its hardware count.
- Check cover removal, fan clearance and LED mounting on the build. Repeat cooling checks with the enclosure installed. Record firmware and companion versions separately.

Source links and design credits.

Primary references researched 03 October 2026. Candidate identities do not prove the purchased SKU.

Rainbow on a Matchstick - Blamm / official design

ARCTIC P9 PWM PST Black - ACFAN00298A

Optional ARCTIC P12 Slim PWM PST - ACFAN00187A

CrowPanel Advanced 5-inch ESP32-P4 - candidate DHE04005D

Mosfet.Party - EPS 8-pin to dual Micro-Fit

Mosfet.Party - PS_ON PC-case-header variant

Mosfet.Party - PS_ON design

3D-Fuel Natural Pro PCTG / material guidance

3D-Fuel print-profile guidance

Prusa INDX 8-tool kit / print volume

CNC Kitchen / PETG clarity experiment, inference for PCTG

MandicReally: source CAD credit for the remixed 92 mm fan shroud, supplied by the owner. Exact original-model URL remains open; a different model is not substituted. Voron manuals informed documentation structure only; no artwork, page assets or prose are reused.

Keep this manual with the CAD revision.

Editable companion data, BOM and open register accompany this PDF. Update figures when the CAD changes.

Source assembly: BC250-Assy.SLDASM

Export: 83 active instances / 34 unique paths / current native CAD tessellation.

Export SHA256: 6629e8ec61feadab36cd367e2baddc19778b2425403b07549a4b3b1a90aa8ac1

Saved assembly observation: 03 October 2026, 21:11:25 UTC; observed saved-file SHA256:

9373a60ab27ae99628ebbc556cbeacaa1fa84fe42846cb7406cfa16978e2c861

Rev 0.3 / design and build preparation

This edition covers the exported CAD inventory; procurement and mechanical interfaces remain unfinished. It includes original lavender/sakura artwork, CAD locator views, LED mount and harness concepts, print trials and a revision register. The owner-proposed sequence starts on page 31. Final exploded views, physical fit validation, procurement and newly flashed features are still pending.

The repository CAD differs from this snapshot. Check current files and open issues for your build or remix; questions are welcome. When the project changes, update the affected BOM rows and links, record the CAD revision and refresh the diagram. Add assembly or test steps after verification. Keep unresolved decisions in the open register.

Build it once. Keep access open.

This order comes from the owner's build plan. It will become the final assembly chapter after CAD and a physical build verify it.

Prepare on the bench	Assemble the machine	Close and commission
01 Test BC250 and prepare BIOS	05 Backbone and mounts	13 Power adapters and wiring
02 Install Linux on direct NVMe	06 Aprons, speakers and display	14 Relocate storage and add Wi-Fi
03 Prepare display and SD media	07 Fans and shroud	15 USB, covers and cable retention
04 Inserts and magnets	08 PSU and IEC adapter	16 First boot at stock settings
	09-11 Heatsink conversion and APU cooler	Then: optional tuning, one change at a time
	12 Matchsticks and their harnesses	

Why this order?

Test the board before building around it. Install the display cables before the aprons close around the glass. Thread the fan cables before fastening the shroud. Leave cable access until the final cover check.

How to use the draft

Follow the order for planning. A step with an unresolved cut, pinout, fastener or mount needs its listed check before use. The appendix keeps proposed hardware separate from the 83 modeled instances.

01 / Test the board before the case

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Use the elektrikM BC250 guides for the initial check. On the bench, connect the required power and display connections with no NVMe installed. Fit the stock cooling components and working temporary fans before first power.
- 2 Make a brief first boot, record the BIOS version and confirm the board reaches firmware setup. Shut down after recording it. Keep cooling running throughout every powered test and BIOS flash.
- 3 Prepare the community BIOS procedure for this exact board. Record the selected ROM, its hash, the stock backup and a recovery method before starting.
- 4 Follow the selected guide and its matching toolkit. The linked kit's Flash.nsh expects Robin5.00. Its bundled stock image is not automatically the modified P3.00 ROM. Do not combine filenames from another flashing script.

Check before moving on

A successful initial POST and a recorded BIOS version come before case assembly. Keep cooling active. The evidence does not establish a safe uncooled boot interval.

Still needs confirmation

Board revision, owner-selected release, original backup and recovery procedure need recording. No BIOS has been flashed as part of this documentation.

[BC250 initial setup and BIOS flashing guide](#)

[Original community BIOS project](#)

02 / Install Linux with direct storage

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Install the intended OS NVMe directly in the BC250 for this bench stage. Prepare the current Bazzite-Deck / Steam Gaming Mode installer using the linked guides. Keep the display microSD card separate from this USB installer.
- 2 Boot the installer normally from the firmware boot menu. If the board drops to an EFI shell, use its mapping information to identify the actual USB filesystem and boot file. USB mappings vary. Do not assume one universal BLK0 or FS0 alias.
- 3 Install to the intended NVMe and confirm it boots. Follow current kernel, Mesa and firmware guidance. The old-version Flatpak Mesa override on the wiki is not a blanket instruction for a current installation.
- 4 Shut down completely and disconnect the bench connections. Keep this same OS drive for the later relocation step. This sequence does not call for an extra SSD.

Check before moving on

The BC250 supports several Linux distributions. Bazzite and Arch-based options are our preferred starting points because the community has dedicated guides and tooling for them. Bazzite is the worked example here.

Still needs confirmation

Other distributions need compatible kernel, Mesa, firmware and governor setup. This is a project preference, not a measured ranking or a promise that every Linux image works unchanged.

Owner-selected Bazzite guide
Community distribution choices

03 / Prepare the display on the bench

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Use the display's matching firmware source and media revision. Compile and check a release before installing it. New LED, action-panel, gamepad and agent features are planned work. They are not all present in the current firmware.
- 2 Prepare a 16-32 GB or larger microSD card, subject to the selected card/filesystem requirements. Put the matching animation and audio assets in a BC250 directory at the card root.
- 3 The current V3 player expects day3, nite3, sleep3 and wake3 assets: .rvj animation files and .pcm audio files. These are firmware-specific media. Copying an ordinary MP4 into that folder does not replace them.
- 4 Check animation, sound and serial resource readings on the bench, then shut down and set the display aside. The existing UART0 resource-monitor link has shown host values. A new firmware release has not yet been flashed.

Check before moving on

Current firmware reads the card at `/sdcard/BC250`. Keep all four scene pairs from one compatible media release. Preserve the user's card contents. The source player does not format a failed card mount.

Still needs confirmation

A reproducible firmware build, release checksum, card filesystem and complete file manifest still need issuing. Confirm the actual board model and speaker ratings before wiring.

04 / Install inserts, then magnets

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Fit four heat-set inserts to each of the two printed 92 mm fan inserts: eight total. Test the selected insert and PCTG pocket on a coupon before using the finished parts.
- 2 Reserve four more inserts for the printed fan-filter mounts. These mounts are not designed yet, so their insert holes and installation details remain pending.
- 3 Install the 6 x 3 magnets in the printed parts, followed by the 10 x 2 magnets. Use the final pocket and retention method once they are tested. The current modeled counts are 14 and 6 respectively.
- 4 For magnet-to-magnet joints, pair the actual mating magnets and mark their orientation. For a cover that meets a steel screw head, check attraction and seating. A steel screw head does not identify a magnet's polarity.

Check before moving on

All inserts must seat correctly without distorting their pockets. Trial-fit each magnetic joint before trapping a magnet in a finished part.

Still needs confirmation

Extra magnet locations, final quantities, adhesive or mechanical retention, grade and supplier part numbers remain open. The four filter inserts are additional proposed hardware.

05 / Prepare the backbone

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 At the three BC250 mounting holes, prepare the proposed 6 mm spacer joints. The owner's plan uses countersunk bolts from the rear and board screws from the front.
- 2 Resolve that joint before fastening: the current CAD spacer reference is unthreaded nylon. Use a verified through-bolt stack or a correctly selected threaded standoff. Two screws cannot independently fasten into an unthreaded spacer.
- 3 Fit six threaded magnets behind the backbone with countersunk fasteners from the front. Confirm magnet body size, actual thread and screw length against the purchased parts.
- 4 Install the future fan-filter mounts at the bottom of the backbone once their geometry and hardware are finalized.

Check before moving on

Check that front-facing heads sit flush and that no fastener enters board clearance. Trial-fit a rear-cover magnet against its intended threaded magnet.

Still needs confirmation

The spacer stack, rear bolt lengths, threaded-magnet part number and filter mounts remain unresolved. No torque is specified before those joints are validated.

06 / Let the aprons hold the display

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Install a speaker in each side apron. Fit the power button in the left apron and the USB hub in the right apron.
- 2 Connect the selected right-angle USB cable to the display's UART0 port. Prepare the LED-control harness at the display only after its connector and pinout are verified.
- 3 Slide the display into the left apron's tabs. Keep the speaker cable behind it. Hold the display while sliding the left apron into the backbone. The proposed fit leaves the front glass slightly proud of the front opening.
- 4 Slide in the right apron, guiding the display into its channels and corner tabs. Keep the right speaker cable behind the display. From the rear, connect both speaker cables to the verified display connectors.

Check before moving on

Support the glass until both aprons retain it. Check that every corner is seated and that no cable is pinched behind the display.

Still needs confirmation

Confirm actual retention clearance, cable bend direction and speaker connectors. The owner's LED connector was unspecified. No display GPIO or pin assignment is issued here.

07 / Fit the fans and thread the cables

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Turn the shroud upside down and place the two 92 mm fans inside it. The owner's airflow direction exhausts through the side of the shroud that the fan frames contact. Confirm the fans' airflow arrows against the final diagram.
- 2 Place one printed fan insert over each fan. Support the backbone upside down or on its side, then slide the shroud into place.
- 3 As the shroud enters, guide the USB-hub cable, power cable, LED-control cable, display USB cable and both fan cables through the rear opening. Keep all cables away from fan blades and screw paths.
- 4 Secure the assembly in all eight holes with the proposed eight M3 x 45 countersunk screws, after checking head geometry, engagement and clearance in the real stack.

Check before moving on

Turn each fan by hand with power disconnected. Check cable clearance, the eight insert engagements and the intended intake-to-heatsink airflow path.

Still needs confirmation

The replacement countersunk screw part number and filter interfaces remain open. The shroud remix is credited to MandicReally. Its exact original-model URL is still needed.

08 / Install the PSU and IEC elbow

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Return the backbone to its working orientation and slide the Apevia 500 W SFX PSU into the rear. The owner's proposed orientation places its fan away from the backbone, with the AC inlet and power switch facing down.
- 2 Secure it in the four mounting positions using screws that match the actual PSU thread. The owner suggested M3 x 8 countersunk screws or the supplied screws. M3 is a candidate, not a confirmed PSU thread.
- 3 With AC disconnected, fit the selected 90-degree IEC adapter so its cable exit faces the rear. Check its connector type, rating, bend direction and cover clearance before turning the assembly over.
- 4 Leave the DC cables accessible for routing. Bundle excess PSU length later, after every connection and cover-removal path is checked.

Check before moving on

The PSU is held securely without an overlong screw entering it. Its fan opening, switch and AC connection remain accessible.

Still needs confirmation

Exact PSU SKU, screw thread and IEC-adapter part number are pending. Do not force a candidate screw into the PSU or modify its supplied mains wiring.

09 / Prepare the heatsink conversion

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Use the exact MandicReally conversion instructions and matching cooler adapter before cutting. Record the contact map and take clear photos of both PCB faces and the original pad locations.
- 2 Remove the eight stock heatsink screws and retain them with their springs identified. Remove the stock heatsink from the PCB before any metal cutting.
- 3 Clean the original paste and pad residue with the owner-specified 99% IPA and a soft microfiber cloth, using a method suitable for the board. Let everything dry fully.
- 4 The owner proposes dividing the heatsink into three sections around the central APU heatpipe section, using a metal-cutting bandsaw in skilled hands. Confirm the cut lines and heatpipe clearance first. Deburr and remove all metal debris before bringing any section back to the PCB.

Check before moving on

Record which components originally contacted pads. Keep the PCB away from cutting and cleaning debris. Every cut section must be clean and free of sharp edges.

Still needs confirmation

The exact creator conversion GitHub/video, cut drawing and fastener stack have not been confirmed. This page records the proposed operation. It does not issue an approved cut plan.

10 / Refit the front heatsink sections

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Use the recorded original contact map to identify the front-side components on either side of the APU. Keep the APU paste area clear of putty.
- 2 The owner's UTP-6 starting deposit is approximately 4 mm high before compression. It is a starting amount, not the intended final chip-to-heatsink gap.
- 3 Press the two smaller outside heatsink sections gently onto their intended contacts while aligning the screw holes. Check the compressed contact pattern rather than using screw force to overcome a poor fit.
- 4 The proposed stack reuses four original non-spring heatsink screws, with four M3 washers behind the PCB. Verify actual engagement, washer footprint and board clearance before securing it.

Check before moving on

Inspect contact coverage and squeeze-out. Confirm the sections sit as intended without board bowing or contact with nearby components.

Still needs confirmation

The compressed gaps and putty contact map need physical verification. Four reused screws are a subset of the retained eight, not a new purchase. The four-washer proposal still needs mapping to the three modeled washers.

11 / Cool the rear VRAM and APU

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 On each actual rear VRAM module, the owner proposes a UTP-6 ball approximately 6 mm tall before compression. The deposit height is not a final gap specification.
- 2 Place the original center heatsink section against the rear VRAM and secure it through the cooler adapter with the correctly selected M3 hardware. Check all chip contacts and neighboring clearances.
- 3 Apply APU thermal paste. Arctic MX-6 is the owner's recommendation. Fit the selected APU cooler with its matching stainless mounting hardware, orienting the fins for the case's airflow.
- 4 After confirming the complete cooler stack, mount the BC250 at the three 6 mm spacer positions. The proposed board screws are three M3 x 8, subject to the spacer-joint check on stage 05.

Check before moving on

Verify paste contact, rear VRAM contact, adapter fit and PCB flatness. The cooler must not depend on cable pressure or contact with the covers.

Still needs confirmation

Rear-sink screw quantity/length, exact cooler revision, adapter subparts and final thermal gaps remain open. Putty and paste quantities are not yet a released purchasing specification.

12 / Connect fans and mount the LEDs

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Connect the two 92 mm fan leads and the single proposed 120 mm fan to the selected three-to-one splitter. Route the leads through the large backbone opening, then connect the splitter to CPU_FAN only after checking its electrical limits.
- 2 Mount the two Rainbow on a Matchstick boards and diffusers to their printed LED mounts using verified M3 hardware. Bolt the mounts to the backbone at the owner's chosen CAD locations.
- 3 Route the display-to-first-bar cable through the large left-side opening. The intended first board is the left Matchstick. The owner describes its input connection at the top.
- 4 Route the inter-bar cable from the left board under the PCB, around the lower edge of the rear VRAM heatsink, to the right board. Check actual DIN/DOOUT labels against page 22. Physical top or bottom does not prove signal direction.

Check before moving on

Only the two ten-LED Matchsticks supply auxiliary RGB. Verify 5 V, ground and data at both ends before power. The reference connectors are not a straight-through three-wire cable.

Still needs confirmation

LED mounts, display pinout, wire sizes, power budget and harness lengths remain open. If the third fan is a P12 Slim, all three fans have a combined listed running current of 0.36 A. Startup current and CPU_FAN capacity are still unverified.

Blamm / Rainbow on a Matchstick design

13 / Verify power before connecting it

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Select the exact Mosfet.Party PS_ON adapter and EPS power-adapter revisions. Follow the installation instructions for those purchased boards.
- 2 The owner proposes extending the PS_ON ground lead by cutting and soldering an extension, routing it through the same left opening and beneath the PCB to the identified ground pin. Confirm that modification, pin and insulation before making it.
- 3 Provide proper insulation and strain relief. The suggested small amount of hot glue is retention only. It does not verify a pinout or replace an electrically sound connection.
- 4 Route the PSU PCIe and CPU EPS leads through the large left opening and connect each to its correct destination. They are distinct connections. The EPS adapter supplements the onboard PCIe power. Swapping the leads can damage the board.

Check before moving on

With AC disconnected, compare connector labels, purchased-adapter instructions and the actual wiring. Check continuity and clearance before any powered test.

Still needs confirmation

No PS_ON variant, modified ground-pin connection or EPS-board revision is approved by this draft. Do not infer compatibility from similar connector housings.

Mosfet.Party BC250 hardware and installation references

14 / Relocate the NVMe and add Wi-Fi

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Move the same OS NVMe from the direct board slot to the selected extension assembly. The proposed extension plate uses four M3 screws on its mount. The mount uses four more M3 screws at the backbone bottom.
- 2 Leave the mount temporarily loose while arranging the cable, with the machine unpowered. Final tightening happens before closure and operation.
- 3 Attach the correct antenna leads to the Wi-Fi 7 M.2 board. Install the selected splitter into the BC250 M.2 slot. The owner proposes the Wi-Fi board in the rear slot near the retaining screw.
- 4 Route the NVMe extension along the rear backbone, through a service loop and the smaller opening beside the BC250, then over the Wi-Fi adapter to the remaining slot. Use a bend radius allowed by the cable. The proposed 90-degree change of direction is not an instruction to crease it.

Check before moving on

Confirm slot keying, lane support and the actual Wi-Fi/NVMe topology before tightening. Check cable retention and the cooler airflow path.

Still needs confirmation

Exact splitter, Wi-Fi board, extension cable and mount geometry remain open. The three modeled SSD instances do not mean three OS drives are required by this sequence.

15 / Finish routing before the covers

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Trial-fit the front magnetic cover against its intended screw heads. Check seating and attraction, then remove it again if it would obstruct cable work.
- 2 Route the display and hub USB connections along the rear. Use the chosen extensions or 180-degree adapters as needed. The owner prefers a blue USB 3.1 host port for the display and a black USB 2.0 port for the hub. Verify actual port labels and connector fit.
- 3 Fit the antenna connectors through the rear cover. Secure excess PSU wiring and the smaller USB/audio/LED cables at verified tie points. Keep the fan-to-heatsink-to-top airflow path clear.
- 4 Tighten the NVMe mount and check every cable and fan. Attach the rear cover to the six threaded magnets, checking mating polarity. Fit the top cover to the backbone/front cover and confirm that all covers remove without pulling on a cable.

Check before moving on

A cover should close without acting as a cable clamp. Remove and refit each cover once. Check fans rotate freely and connectors retain enough service slack.

Still needs confirmation

Tie locations, connector angles and final cable lengths still need a physical trial and CAD updates. Steel screw-head attraction and magnet-to-magnet polarity are different checks.

16 / Commission at stock settings

Owner-proposed sequence. Confirm the details in the right-hand boxes before doing the work.

- 1 Connect keyboard, mouse and DisplayPort monitor. Use Ethernet for first startup if available. Check the cooling connections, then start the machine at stock settings.
- 2 Confirm Linux boots, configure Wi-Fi and check storage and USB devices. Install BC250 Control Center using the current release instructions for the selected distribution.
- 3 Connect the Ronin host resource-monitor service to the display and compare its readings with the host. The Windows bench monitor has produced readings. Linux hardware validation and finished companion installers remain pending.
- 4 Record idle and load behavior with the covers fitted. Only after the stock build is stable should optional core activation, overclocking or fan-curve changes be considered. Change one setting at a time and retain a recovery path.

Check before moving on

Check fan operation and temperatures under load, plus storage, Wi-Fi, display, speakers and USB. Record the software/BIOS versions with the results.

Still needs confirmation

This draft supplies no promised overclock or performance result. Gamepad wake, advanced ARGB, context-sensitive action buttons and the AI companion portal still need firmware and host integration.

Build-plan hardware 1 / 8

Owner quantities below are proposals. They are not final required quantities or instructions to order replacements.

Proposed item	Owner qty	Related CAD baseline	Confirm before release
Fan-insert heat-set inserts	8	See CSV for related modeled parts	Eight proposed, four per fan insert. Select the insert dimensions and validate the PCTG pocket.
Future fan-filter mount heat-set inserts	4	See CSV for related modeled parts	Four additional inserts reserved for the future filter mounts. Their holes and thread are not designed yet.
Printed fan-filter mounts	TBD	No named CAD match	Design the filter mounts and select the filter before fixing mount quantity or hardware.
6 x 3 magnets in printed parts	TBD	See CSV for related modeled parts	Current CAD has 14. More locations are planned. Final count, grade and retention need checking.
10 x 2 magnets in printed parts	TBD	See CSV for related modeled parts	Current CAD has 6. Verify the pockets, mating parts, retention and selected supplier.
Threaded magnets on backbone	6	See CSV for related modeled parts	Six behind the backbone. Confirm actual M3 thread, body size and rear-cover mating orientation.

The frozen inventory remains 83 instances / 34 CAD paths. The complete proposed-hardware CSV/JSON contains 59 rows, including context and reused parts. Blank final quantities and purchase part numbers mean unresolved.

Build-plan hardware 2 / 8

Owner quantities below are proposals. They are not final required quantities or instructions to order replacements.

Proposed item	Owner qty	Related CAD baseline	Confirm before release
Countersunk bolts for threaded magnets	TBD	No named CAD match	Select count, thread, length and countersunk head against the six actual magnet joints.
6 mm BC250 spacers	3	See CSV for related modeled parts	Three positions, 6 mm long. Current modeled spacers are unthreaded nylon. Resolve the two-sided screw stack.
Rear countersunk bolts for BC250 spacer joint	TBD	See CSV for related modeled parts	Confirm the rear bolts and their engagement with the selected spacer joint. Existing bolts and nuts may be reused.
BC250-to-spacer screws	3	See CSV for related modeled parts	Three M3 x 8 proposed at the board. Verify engagement, insulation and clearance after selecting the spacer joint.
Fan/shroud countersunk screws	8	See CSV for related modeled parts	Eight M3 x 45 match the modeled count and nominal length. Select the countersunk head and check the assembled stack.
PSU mounting screws: M3x8 candidate OR supplied screws	4	See CSV for related modeled parts	Four screws total. Use the actual PSU thread. M3 x 8 and supplied screws are alternatives, not eight screws.

The frozen inventory remains 83 instances / 34 CAD paths. The complete proposed-hardware CSV/JSON contains 59 rows, including context and reused parts. Blank final quantities and purchase part numbers mean unresolved.

Build-plan hardware 3 / 8

Owner quantities below are proposals. They are not final required quantities or instructions to order replacements.

Proposed item	Owner qty	Related CAD baseline	Confirm before release
90 degree IEC adapter	1	No named CAD match	One right-angle adapter. Confirm IEC type, electrical rating, bend direction and rear-cover clearance.
120 mm fan	1	See CSV for related modeled parts	One 120 mm fan in the proposed sequence. Exact model and thickness remain open. Do not automatically add a second cooler fan.
Three-to-one fan splitter	1	See CSV for related modeled parts	One three-output splitter. Check total running/startup current, CPU_FAN limit, PWM wiring and a single tach signal.
Original stock heatsink screws removed and retained	8	See CSV for related modeled parts	Keep all eight original screws and identify their springs. These retained parts are not a new purchase.
Original screws reused for outer heatsink sections	4	See CSV for related modeled parts	Reuse four of the retained eight for the outside heatsink sections. Confirm the exact screw stack.
M3 washers behind PCB for outer sink sections	4	See CSV for related modeled parts	Four washers proposed. CAD currently has three washers of unresolved role. Compare joints before deciding what to buy.

The frozen inventory remains 83 instances / 34 CAD paths. The complete proposed-hardware CSV/JSON contains 59 rows, including context and reused parts. Blank final quantities and purchase part numbers mean unresolved.

Build-plan hardware 4 / 8

Owner quantities below are proposals. They are not final required quantities or instructions to order replacements.

Proposed item	Owner qty	Related CAD baseline	Confirm before release
M3 screws for rear center heatsink through cooler adapter	TBD	See CSV for related modeled parts	Choose quantity, length and head after checking the rear heatsink and cooler adapter. No complete screw stack is confirmed.
Cooler's stainless mounting hardware	TBD	See CSV for related modeled parts	Use the mounting kit that matches the selected cooler and adapter. Confirm each physical part in the imported adapter assembly.
Upsiren UTP-6 thermal putty: front components	TBD	No named CAD match	UTP-6, about 4 mm initial deposit height before compression. Verify the contact map and final gap. Mass remains unknown.
Upsiren UTP-6 thermal putty: rear VRAM	TBD	No named CAD match	UTP-6, about 6 mm initial ball height per rear VRAM module before compression. Confirm coverage and final gap.
Arctic MX-6 thermal paste	TBD	No named CAD match	Arctic MX-6 is the recommended APU paste. Determine the amount from the verified cooler/contact procedure.
Display microSD card	1	No named CAD match	One display card, 16-32 GB or larger proposed. Confirm capacity, filesystem, speed and matching firmware media.

The frozen inventory remains 83 instances / 34 CAD paths. The complete proposed-hardware CSV/JSON contains 59 rows, including context and reused parts. Blank final quantities and purchase part numbers mean unresolved.

Build-plan hardware 5 / 8

Owner quantities below are proposals. They are not final required quantities or instructions to order replacements.

Proposed item	Owner qty	Related CAD baseline	Confirm before release
90 degree USB cable to display UART0	1	See CSV for related modeled parts	One right-angle cable at UART0. Check connector type, exit direction, data support and length.
LED mounts/brackets	TBD	No named CAD match	Mounts are not modeled yet. Confirm PCB-hole dimensions, diffuser clearance, removal access and bracket quantity.
M3 screws for Matchstick/diffuser to mounts	TBD	See CSV for related modeled parts	M3 hardware proposed for board/diffuser mounting. Select quantity and length after the LED mounts are designed.
Bolts for LED mounts to backbone	TBD	No named CAD match	Choose bracket-to-backbone fasteners after checking the available holes, countersinks and electrical clearance.
Display-to-first Matchstick control harness	1	No named CAD match	One display-to-left-board harness. Display connector, GPIO, logic level, wire size and power source remain open.
First-to-second Matchstick bridge harness	1	No named CAD match	One left-to-right bridge. Verify DIN/DOUT mapping and the measured route around the rear VRAM heatsink.

The frozen inventory remains 83 instances / 34 CAD paths. The complete proposed-hardware CSV/JSON contains 59 rows, including context and reused parts. Blank final quantities and purchase part numbers mean unresolved.

Build-plan hardware 6 / 8

Owner quantities below are proposals. They are not final required quantities or instructions to order replacements.

Proposed item	Owner qty	Related CAD baseline	Confirm before release
Mosfet.Party PS_ON adapter	1	No named CAD match	One Mosfet.Party adapter. Confirm the purchased variant and approved ground connection before extending its lead.
Mosfet.Party power adapter (EPS intent)	1	No named CAD match	One Mosfet.Party EPS power adapter. Verify the purchased revision and distinct EPS/PCIe connections.
PS_ON ground extension wire and insulation/strain relief	TBD	No named CAD match	Extension length, gauge, insulation and strain relief depend on the selected adapter and verified route.
NVMe extension cable/assembly	1	No named CAD match	One extension assembly. Confirm electrical compatibility, connector fit, length and minimum bend radius.
NVMe extension plate	1	No named CAD match	One plate proposed. Confirm its actual design and SSD/extension mounting interfaces.
NVMe mount	1	No named CAD match	One bottom mount proposed. Add it to CAD and verify airflow, cable access and cover clearance.

The frozen inventory remains 83 instances / 34 CAD paths. The complete proposed-hardware CSV/JSON contains 59 rows, including context and reused parts. Blank final quantities and purchase part numbers mean unresolved.

Build-plan hardware 7 / 8

Owner quantities below are proposals. They are not final required quantities or instructions to order replacements.

Proposed item	Owner qty	Related CAD baseline	Confirm before release
M3 screws: extension plate to NVMe mount	4	No named CAD match	Four M3 screws, plate to mount. Length and head depend on the final printed stack.
M3 screws: NVMe mount to backbone	4	No named CAD match	Four M3 screws, mount to backbone. Arrange wiring while loose, then secure before operation.
Same OS NVMe reused after installation	1	See CSV for related modeled parts	The same OS drive moves from direct installation to the extension. It is not an additional drive order.
Wi-Fi 7 M.2 board	1	No named CAD match	One Wi-Fi 7 board proposed. Exact radio, keying, driver support and splitter compatibility need documenting.
USB extension cables (alternative)	TBD	No named CAD match	Choose extensions or compatible 180-degree adapters as the route requires. Do not sum both alternatives automatically.
180 degree USB adapters (alternative)	TBD	See CSV for related modeled parts	Alternative to USB extensions where suitable. Confirm connector orientation, data support and clearance.

The frozen inventory remains 83 instances / 34 CAD paths. The complete proposed-hardware CSV/JSON contains 59 rows, including context and reused parts. Blank final quantities and purchase part numbers mean unresolved.

Build-plan hardware 8 / 8

Owner quantities below are proposals. They are not final required quantities or instructions to order replacements.

Proposed item	Owner qty	Related CAD baseline	Confirm before release
Zip ties	TBD	No named CAD match	Count and size follow the real cable bundles. Keep fans, vents and cover-removal paths clear.
Cable anchors/tie points/harness retention	TBD	No named CAD match	Design tie points and strain relief after routing the actual cables. Cover panels should not support heavy PSU bundles.
Fan filter/media	TBD	No named CAD match	Select filter size/material and service access together with the future printed mounts.

The frozen inventory remains 83 instances / 34 CAD paths. The complete proposed-hardware CSV/JSON contains 59 rows, including context and reused parts. Blank final quantities and purchase part numbers mean unresolved.

What this revision actually confirms.

The owner supplies the intended order. Linked project sources inform the checks. They do not certify this physical build.

Evidence	Status in this edition
Owner plan and clarification	Assembly intent and proposed hardware are recorded in the companion data. The 4 mm / 6 mm UTP-6 deposits are starting amounts before compression.
BIOS and installer references	Owner-selected wiki and original BIOS project checked. The candidate modified ROM hash is recorded. No new board backup or BIOS flash was performed for this documentation.
Linux choice	Bazzite is the worked installation example. Other compatible distributions remain options. Bazzite and Arch-based choices have dedicated community guidance.
Creator conversion and hardware	MandicReally credit retained. Exact conversion cut plan, selected cooler stack, PS_ON variant, harness pinout and fan-header budget remain open.
Proposed quantities	Eight heat-set inserts for the fans plus four for the filter mounts are proposed allocations. Four reused screws are within the original eight. PSU and USB alternatives are not added together.
Release checks	Final exploded views and installation details follow completed CAD and physical verification. Existing working display behavior stays distinct from planned functions.

See the linked references in this manual. Proposed hardware: BC250_Proposed_Hardware_Rev_0p3.json. The 21-item original open register is preserved; additional build-plan checks add to it.